

Notes: Holmes (JPE, 1998)

The Effect of State Policies on the Location of Manufacturing: Evidence from State Borders

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1 Big picture

- **Research question.** Do state-level probusiness policies affect where manufacturing activity locates within the United States?
- **Main idea.** Comparing all right-to-work states to all non-right-to-work states is confounded by geography, history, climate, agricultural decline, union culture, transportation, and many other state-level differences. Holmes instead asks what happens exactly at state borders where policy changes.
- **Policy proxy.** A state is classified as **probusiness** if it has a right-to-work law and **antibusiness** if it does not. Right-to-work status is not treated as the only policy that matters; it is used as a marker for a broader policy bundle that tends to be more favorable to manufacturing.
- **Empirical object.** The main outcome is county manufacturing employment as a share of total private nonagricultural employment. The second outcome is county manufacturing employment growth from 1947 to 1992.
- **Identification idea.** If nonpolicy geography changes smoothly at borders, but state policy changes discretely, then a discrete jump in manufacturing activity at the border is evidence that policy matters.
- **Main finding.** Manufacturing activity jumps sharply when one crosses from an antibusiness state into a probusiness state. In the main regression, the 1992 manufacturing employment share rises by about 6.6 percentage points, roughly one-third of the average manufacturing share.
- **Growth evidence.** Manufacturing employment growth from 1947 to 1992 is also higher on the probusiness side of the border, with the main model estimating a 23.1-point increase in the growth measure.
- **Extensions.** A broader Fantus business-climate ranking also predicts border discontinuities, including at borders where right-to-work status does not change. Probusiness policies also raise the share of employment in large manufacturing establishments.

Economic takeaway. State policies appear to have real effects on the spatial allocation of manufacturing. The paper is important because it turns a broad cross-state policy question into a border-comparison design that is much closer to a local experiment.

2 Incremental contribution of the paper

- **Moves beyond cross-state correlations.** Earlier evidence that right-to-work states grew faster in manufacturing is hard to interpret because those states differ systematically from non-right-to-work states. Holmes focuses on border discontinuities, where nearby counties are more comparable.
- **Introduces a geographic discontinuity design.** The key empirical move is to transform a policy question into a comparison of adjacent counties on opposite sides of state borders.
- **Uses the border as a control for geography.** Climate, soil, access to rivers, transportation, agglomeration, and regional demand should change relatively smoothly over space. State policy changes abruptly at the border.

- **Separates levels and growth.** The paper studies both the level of manufacturing concentration in 1992 and postwar growth in manufacturing employment.
- **Builds a spatial regression model.** Holmes allows the expected manufacturing share to vary flexibly along the border and linearly with distance from the border, while estimating a discrete policy shift at the border.
- **Examines multiple robustness checks.** The paper excludes the coal region, removes the West in the baseline, includes the West as a check, studies earlier census years, and estimates placebo discontinuities at simulated borders.
- **Expands beyond right-to-work laws.** The Fantus ranking exercise shows that the border effect is not only about right-to-work status; broader probusiness policies are correlated with manufacturing activity.
- **Highlights local interpretation.** The estimated border effect should not be read as a statewide average treatment effect or as the causal effect of one specific statute. It is the local effect of a policy bundle near borders.

3 Data and measurement

3.1 State policy classification

The baseline classification is simple. A state with a right-to-work law is coded as probusiness, and a state without such a law is coded as antibusiness. A right-to-work law bans union-shop arrangements in which all employees are required to join the union.

- **Why this variable?** Right-to-work laws plausibly matter to manufacturers because manufacturing is relatively union-sensitive.
- **Why it is also a proxy.** States that pass right-to-work laws tend to adopt other policies favorable to manufacturing, including lower taxes on capital, weaker regulation, and subsidies for plant location.
- **Important interpretation.** The coefficient is not the effect of passing a right-to-work law while holding all other policies fixed. It is the effect of crossing into the side of the border with the right-to-work/probusiness policy bundle.

3.2 Geographic unit and border construction

The unit of observation is the county. Counties are used because they are the finest national geographic unit with comprehensive Census Bureau data.

The **policy change border** is the set of state borders separating right-to-work states from non-right-to-work states. In the main analysis, Holmes excludes western states because western counties are very large, sparsely populated, and geographically coarse. This leaves two main border segments:

- **Segment 1: Confederacy border.** This segment runs from the western end of the Oklahoma-Texas border to the Maryland-Virginia border at the Atlantic Ocean.

- **Segment 2: Plains/Midwest border.** This segment runs from the Minnesota-North Dakota border at Canada to the western end of the Oklahoma-Kansas border.

Each county is mapped into a border coordinate system. Let $mindist_i$ be the distance from county i 's population centroid to the nearest point on the policy change border. Define

$$y_i = \begin{cases} mindist_i, & \text{if county } i \text{ is in an antibusiness state,} \\ -mindist_i, & \text{if county } i \text{ is in a probusiness state.} \end{cases} \quad (1)$$

The variable x_i is the mile marker on the border closest to the county. This converts the complicated U.S. map into a two-dimensional space: x moves along the border, and y moves across the border.

3.3 Manufacturing outcomes

The main level outcome is:

$$\text{Manufacturing share}_{i,1992} = \frac{\text{manufacturing employment}_{i,1992}}{\text{total private nonagricultural employment}_{i,1992}}. \quad (2)$$

The second outcome is the 1947–92 manufacturing employment growth rate:

$$growth_i = 100 \times \frac{emp_{i,92} - emp_{i,47}}{0.5emp_{i,47} + 0.5emp_{i,92}}. \quad (3)$$

This growth measure is bounded between -200 and 200 , which avoids infinite growth rates for counties with zero employment in 1947.

3.4 Data sources

- **County Business Patterns.** Used for 1992 manufacturing and total private employment, including establishment size classes.
- **Census of Manufactures.** Used for county manufacturing employment in several years.
- **Census of Population.** Used when employment shares relative to county population are needed for earlier years.
- **Geographic coordinates.** County population centroids are used to measure distance to the policy border.
- **Fantus ranking.** The 1975 Fantus legislative business climate ranking is used as an alternative policy measure.

3.5 Coal region and western states

The coal region near the Kentucky-Virginia and West Virginia-Virginia borders is usually excluded. The reason is that coal geology creates a natural discontinuity at the border: the mining employment share changes sharply for reasons unrelated to state policy.

Western states are also excluded from the baseline because county size is very large and distance to the border is measured imprecisely. Holmes later shows that including western states does not substantially change the estimates.

4 Methodology and empirical specifications

4.1 Border discontinuity intuition

The empirical logic is close to a spatial regression discontinuity design. If policy has no effect, manufacturing activity should vary smoothly when one crosses a state border. If policy matters, manufacturing activity should jump discretely at borders where policy changes.

The key comparison is therefore not South versus North, or right-to-work states versus all other states. It is counties close to the same border, on opposite sides of the state line.

4.2 Theoretical model

Holmes develops a simple model with two jurisdictions, a probusiness South and an antibusiness North. Manufacturing entrepreneurs may remain in place, not produce, or move to an alternative location. The antibusiness policy imposes a cost c in the North; moving costs increase with distance.

The model gives several lessons:

- **A border jump is expected if policy matters.** If the policy cost $c > 0$, manufacturing's share of local employment jumps at the border.
- **The border effect can be larger than the interior effect.** Entrepreneurs close to the border can cheaply move to the probusiness side, so the local border discontinuity may exceed the statewide average effect.
- **The effect may or may not fade with distance.** Depending on moving costs and underlying geography, the effect can fizzle out away from the border or remain large in the interior.
- **Welfare is not identified.** A border jump can reflect new wealth creation or mere relocation. The border design identifies location effects, not welfare effects.

4.3 Cross-tabulations by distance layers

The first empirical exercise groups counties into 25-mile bands on each side of the border. For example:

- antibusiness border layer: $0 < y_i \leq 25$;
- probusiness border layer: $-25 \leq y_i < 0$;
- interior layers: 25–50, 50–75, and 75–100 miles on each side.

Holmes compares average manufacturing shares and growth rates across adjacent layers. The central question is whether the change at the actual border is much larger than changes between interior layers.

4.4 Main spatial regression

The main regression for the 1992 manufacturing share is:

$$share_i = \theta probusinessdum_i + \alpha(x_i) + \beta(x_i)y_i + \epsilon_i. \quad (4)$$

- $probusinessdum_i$ equals one if county i is on the probusiness side.
- θ is the discontinuous border shift.
- $\alpha(x_i)$ allows the expected manufacturing share to vary flexibly along the border.
- $\beta(x_i)y_i$ allows a smooth trend as one moves perpendicular to the border.

Holmes approximates $\alpha(x)$ with a fourth-degree polynomial separately for each of the two border segments. He considers four choices for $\beta(x)$: no trend, one constant trend, separate constant trends by segment, and general fourth-degree trends by segment.

4.5 Growth regression

The same statistical structure is applied to the 1947–92 manufacturing employment growth outcome. The parameter θ then measures the discontinuous jump in postwar manufacturing growth when crossing into the probusiness side.

4.6 Simulated-border placebo test

To test whether any arbitrary line would produce a jump, Holmes creates simulated borders within the same side of the true border. For example, he estimates discontinuities at $y = 50$ and $y = -50$ instead of $y = 0$.

The result is that the large discontinuity appears only at the actual policy border. This supports the interpretation that the border effect is not just an artifact of the spatial model.

4.7 Alternative policy variable: Fantus ranking

The paper then replaces right-to-work status with the 1975 Fantus business climate ranking. The model is:

$$share_i = -\phi Fantus_i + \alpha_{b_i}(x_i) + \beta_{b_i}(x_i)y_i + \epsilon_i. \quad (5)$$

The minus sign appears because a smaller Fantus number means a better business climate. The coefficient ϕ measures how much manufacturing activity changes at state borders when Fantus rankings differ.

This extension uses all borders among the 48 contiguous states, not just borders where right-to-work status changes.

4.8 Large-establishment extension

The final extension studies establishment size. Holmes defines large establishments as those with 100 or more employees. The question is whether probusiness policies are especially associated with large manufacturing establishments.

This matters because large plants are more likely to be unionized and more capital-intensive. If policies weaken unions or lower capital taxes, the effects should be stronger for large manufacturing establishments than for small establishments or nonmanufacturing establishments.

5 Identification and interpretation

5.1 What the design identifies

The paper identifies whether manufacturing activity changes discontinuously when crossing state borders where policy changes. This is a local border effect of a state policy bundle.

It is not a pure estimate of a single law, and it is not automatically the statewide average effect of probusiness policies. A large border effect can arise because firms near the border are especially responsive to small moving costs.

5.2 Why the border comparison helps

The border comparison controls for smooth spatial determinants of manufacturing:

- climate;
- soil fertility and agricultural history;
- access to rivers, coasts, and transportation;
- agglomeration forces;
- broad regional demand;
- cultural attitudes that change gradually over space.

If these factors are smooth at the border, a discontinuity at the border is difficult to explain without a discontinuity in policy or institutions.

5.3 Main threats to interpretation

- **Natural discontinuities.** Geography can sometimes change sharply at state borders. The coal region is the clearest example, and Holmes excludes it.
- **Historical discontinuities.** State borders can also coincide with older institutional or demographic discontinuities unrelated to current business policy.
- **Right-to-work as a proxy.** The right-to-work variable captures a bundle of policies and political attitudes. The estimate should not be interpreted as the isolated effect of one law.

- **Local effect only.** The border effect can be larger than effects far inside a state, because relocating across the border is cheapest near the border.
- **No welfare conclusion.** A relocation of manufacturing toward probusiness states may or may not increase aggregate welfare.

5.4 How the paper addresses concerns

- The coal-region border is excluded from most estimates.
- The paper uses flexible functions of distance along the border, so high-manufacturing parts of the country are compared to nearby high-manufacturing parts, and low-manufacturing parts to nearby low-manufacturing parts.
- Separate analysis of the two border segments shows similar share jumps.
- Table 4 shows that the share difference is widespread across many individual state border pairs.
- Simulated borders do not produce similar discontinuities.
- The 1947 estimates are close to zero, suggesting the discontinuity developed over the postwar period.
- The Fantus ranking produces effects at non-right-to-work borders, supporting the broader policy interpretation.

5.5 Interpretation of the main coefficient

In the main specification, $\theta = 6.6$ for the 1992 manufacturing share. Since the average manufacturing share is about 20 percent, this is economically large: crossing into the probusiness side raises the manufacturing employment share by roughly one-third.

For growth, $\theta = 23.1$, meaning that the bounded 1947–92 manufacturing growth measure is about 23 points higher on the probusiness side in the most general specification.

6 Tables and figures

6.1 Figures 1 and 2: Policy geography and theoretical border effects

Read Figure 1:

- Shaded states have right-to-work laws; unshaded states do not.
- Right-to-work states are concentrated in the South and Plains states, while the traditional manufacturing belt does not have right-to-work laws.
- This geographic clustering is exactly why simple right-to-work versus non-right-to-work comparisons are confounded.

Read Figure 2:

- Panel a shows a border effect that fizzles out away from the border.

- Panel b shows the case with infinite moving costs: the policy creates a permanent level difference between jurisdictions.
- Panel c shows the case with zero moving costs: manufacturing relocates across the border even if the aggregate welfare effect is small.
- Panel d shows that underlying geographic trends can make the spatial pattern look like fizzling even when policy effects away from the border are large.

Economic message: Right-to-work status is geographically nonrandom, but a border discontinuity can still reveal policy effects. The theory also warns that border effects are local and not welfare estimates.

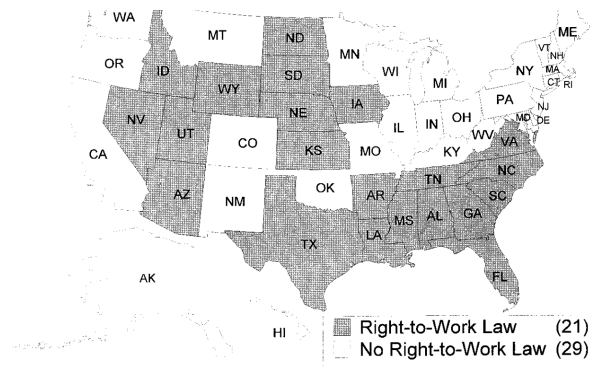


FIG. 1.—Geography of right-to-work laws

Figure 1: Geography of right-to-work laws.

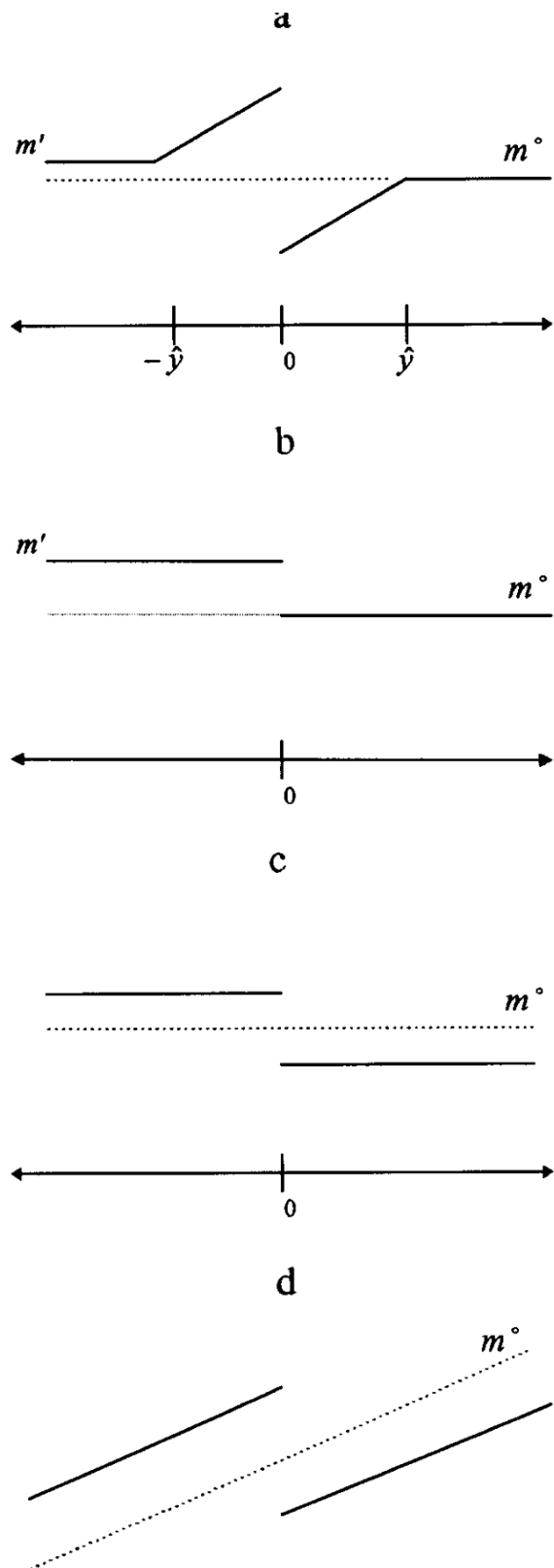


Figure 2: Theoretical border effects.

6.2 Figures 3 and 4: Border sample and manufacturing geography

Read Figure 3:

- Counties within 25 miles of the policy change border are highlighted.
- Dark areas are counties on the probusiness side; light areas are counties on the antibusiness side.
- The figure shows why the main analysis excludes the West: western counties are much larger, so distance to the border is measured less precisely.

Read Figure 4:

- Counties are shaded by deciles of 1992 manufacturing share.
- High manufacturing-share counties are concentrated in the South and parts of the eastern United States.
- Along parts of the policy border, darker counties begin almost exactly at the probusiness side of the border.
- The Kentucky and West Virginia coal region is visually visible as an area where low manufacturing shares reflect mining geography rather than policy.

Economic message: The maps motivate the empirical design. There is suggestive visual evidence of a border discontinuity, but the statistical analysis is needed to average out noise and control for trends along the border.

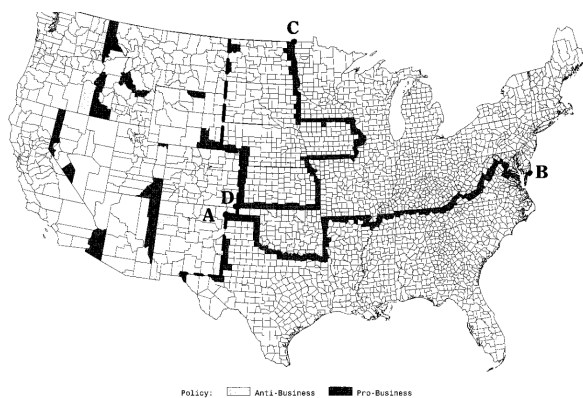


FIG. 3.—Counties within 25 miles of the policy change border

Figure 3: Counties within 25 miles of the policy change border.

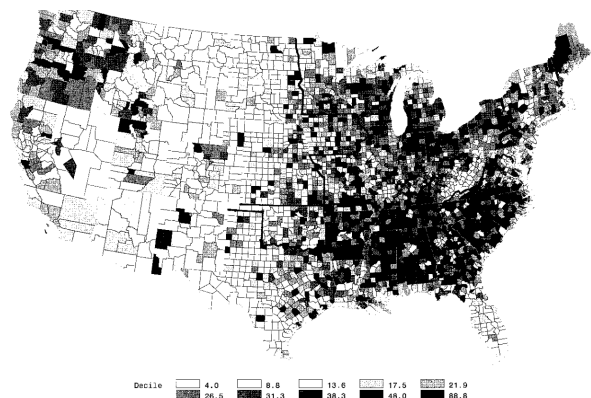


FIG. 4.—Distribution of 1992 manufacturing shares: county deciles

Figure 4: Distribution of 1992 manufacturing shares.

6.3 Tables 1 and 2: Distance layers and adjacent-layer tests

Read Table 1:

- Manufacturing shares are higher in the probusiness border layer than in the antibusiness border layer.
- Excluding the coal region, the share is 27.9 percent on the probusiness side and 22.1 percent on the antibusiness side.

- Growth is also higher: 104.2 on the probusiness side versus 77.2 on the antibusiness side.
- Interior layers do not display changes as large as the jump at the border.

Read Table 2:

- The adjacent-layer test asks whether changes at the actual border are unusually large relative to changes between other neighboring layers.
- The share drop when moving from the probusiness border layer to the antibusiness border layer is 5.8 percentage points, with a p -value of 0.003.
- The growth drop at the border is 27.0 points, with a p -value of 0.008.
- Other adjacent-layer differences are smaller and statistically insignificant.

Economic message: The discontinuity occurs where policy changes, not gradually across all distance bands. This is the first direct evidence that the border itself matters.

TABLE 1
MANUFACTURING EMPLOYMENT SHARES AND GROWTH RATES: CROSS-COUNTY
AVERAGES BY DISTANCE FROM BORDER AND SIDE OF BORDER

MILES FROM BORDER	COAL REGION INCLUDED		COAL REGION EXCLUDED	
	Share of 1992 Total (1)	Growth Rate, 1947–92 (2)	Share of 1992 Total (3)	Growth Rate, 1947–92 (4)
A. Antibusiness Side of Border				
75–100	25.9	67.5	25.0	68.2
50–75	23.1	62.7	25.0	80.9
25–50	23.2	82.0	24.7	88.8
0–25	21.0	62.4	22.1	77.2
B. Probusiness Side of Border				
0–25	28.6	100.7	27.9	104.2
25–50	26.7	89.1	25.5	88.3
50–75	26.7	92.9	24.5	90.1
75–100	25.4	91.8	23.1	93.5

Table 1: Manufacturing shares and growth by distance from border.

TABLE 2
TESTS OF EQUALITY OF MEANS OF ADJACENT LAYERS
(Coal Region Excluded)

ADJACENT COUNTY LAYERS	SHARE		GROWTH RATE	
	Change in Mean (1)	p -Value for Test of Equality (2)	Change in Mean (3)	p -Value for Test of Equality (4)
Anti:50–75 → anti:75–100	.0	.975	12.7	.259
Anti:25–50 → anti:50–75	.3	.880	–7.9	.463
Anti:0–25 → anti:25–50	2.6	.185	11.6	.283
Pro:0–25 → anti:0–25	–5.8	.003	–27.0	.008
Pro:25–50 → pro:0–25	2.4	.217	15.9	.104
Pro:50–75 → pro:25–50	1.0	.620	–1.8	.863
Pro:75–100 → pro:50–75	1.4	.517	–3.4	.742

Table 2: Equality tests for adjacent distance layers.

6.4 Tables 3 and 4: Segment-level and state-pair evidence

Read Table 3:

- Segment 1 is the Confederacy border; segment 2 is the Plains/Midwest border.
- In both segments, the manufacturing share is about 5.5 percentage points higher on the probusiness border layer.
- Growth differences are especially strong for segment 2: 104.0 on the probusiness side versus 54.7 on the antibusiness side.
- Segment 2 also has an underlying trend toward higher manufacturing shares as one moves toward the Midwest industrial states, which motivates controlling flexibly for geography.

Read Table 4:

- The border is broken into 17 individual state-pair comparisons.
- Manufacturing shares are higher on the probusiness side for almost every pair.

- The two exceptions have nearly equal shares, rather than large reversals.
- Growth evidence is noisier, but higher growth on the probusiness side is still widespread.

Economic message: The manufacturing-share effect is not driven by one state pair or one part of the country. It is visible across much of the long policy border.

TABLE 3
MANUFACTURING EMPLOYMENT SHARES AND GROWTH RATES BY SEGMENT AND
DISTANCE FROM BORDER

MILES FROM BORDER	1992 SHARE		1947–92 GROWTH	
	Segment 1: Confederacy Border*	Segment 2: Plains States Border	Segment 1: Confederacy Border*	Segment 2: Plains States Border
	(1)	(2)	(3)	(4)
A. Antibusiness Side of Border				
75–100	25.4	24.4	75.9	58.3
50–75	23.0	26.7	97.7	67.5
25–50	28.5	21.1	101.8	76.5
0–25	26.6	17.7	99.1	54.7
B. Probusiness Side of Border				
0–25	32.3	23.2	104.5	104.0
25–50	30.4	20.3	85.8	91.0
50–75	28.3	19.7	88.8	91.7
75–100	28.5	17.1	97.7	89.1

* Excludes coal region.

Table 3: Manufacturing shares and growth by
border segment.

6.5 Tables 5 and 6: Main statistical model and simulated borders

Read Table 5:

- The main parameter is θ , the discrete shift when crossing into a probusiness state.
- For 1992 manufacturing shares, θ rises from 3.4 with no trend to 6.6 in the most general specification.
- The preferred estimate of 6.6 is economically large and statistically precise.
- For 1947–92 growth, the most general estimate is 23.1.
- Allowing trends to differ by segment matters because the Plains/Midwest border has a strong spatial trend.

Read Table 6:

- Simulated borders are placed at different values of y away from the true border.
- The true border, $y = 0$, produces a much larger estimated jump than any simulated border.
- For the 50-mile sample, the true border share jump is 9.1 and the growth jump is 39.9.
- Placebo borders such as $y = 50$, $y = -50$, and $y = -100$ produce small, statistically insignificant effects.

Economic message: The regression evidence confirms the visual and cross-tab evidence. The discontinuity is special to the actual policy border rather than a generic feature of nearby geography.

TABLE 4
MANUFACTURING EMPLOYMENT SHARES AND GROWTH RATES BY INDIVIDUAL BORDERS

BORDER STATES		1992 SHARE		GROWTH RATE, 1947–92	
Probusiness	Antibusiness	Probusiness (1)	Antibusiness (2)	Probusiness (3)	Antibusiness (4)
Texas	Oklahoma	17.3	16.1	29	54
Arkansas	Oklahoma	43.5	27.6	152	144
Arkansas	Missouri	40.7	30.1	158	125
Tennessee	Missouri	47.8	39.3	100	78
Tennessee	Kentucky	48.4	38.7	142	122
Virginia	Kentucky	17.7	3.4	143	–55
Virginia	West Virginia	31.7	20.1	59	5
Virginia	Maryland	10.3	8.5	89	84
North Dakota	Minnesota	16.2	6.3	137	20
South Dakota	Minnesota	16.2	11.1	138	27
Iowa	Minnesota	28.5	25.1	130	85
Iowa	Wisconsin	29.9	30.2	109	103
Iowa	Illinois	33.6	23.1	75	13
Iowa	Missouri	25.2	16.7	121	122
Nebraska	Missouri	13.8	10.0	69	167
Kansas	Missouri	21.1	22.6	78	96
Kansas	Oklahoma	19.5	11.2	80	–12

Table 4: Manufacturing shares and growth by
individual borders.

TABLE 5
STATISTICAL MODEL

Parameter	Specification 1: No Trend, $\beta(x) = 0$ (1)	Specification 2: Constant Trend, $\beta(x) = \beta_0$ (2)	Specification 3: Different Constant for Each Segment, $\beta(x) \in \{\beta_1, \beta_2\}$ (3)	Specification 4: Trend $\beta(x)$ a General Function of x (4)
A. County Manufacturing Shares* ($N = 951$)				
θ	3.4 (.9)	6.4 (1.6)	6.5 (1.6)	6.6 (1.6)
β_0	...	.03 (.01)	...	...
β_1	...	...	-.01 (.02)	...
β_2	...	...	.08 (.02)	...
R^2	.306	.310	.330	.350
B. County Growth Rates* ($N = 892$)				
θ	19.1 (5.0)	21.2 (9.4)	21.2 (9.4)	23.1 (9.2)
β_0	...	.02 (.09)	...	...
β_1	...	...	.08 (.10)	...
β_2	...	...	-.04 (.10)	...
R^2	.118	.118	.120	.161

* Excludes coal region.

Table 5: Statistical model estimates of the policy-border shift.

TABLE 6

STATISTICAL MODEL: ESTIMATES OF SHIFT PARAMETER θ FOR SIMULATED BORDERS, COUNTIES 50 MILES ABOVE AND BELOW SIMULATED BORDER

Location of Simulated Border	1992 Share	1947-92 Growth Rate
$y = 100$	-.0 (2.7)	-20.9 (15.8)
$y = 75$	1.8 (2.7)	-4.6 (15.3)
$y = 50$	1.1 (2.5)	6.1 (14.4)
$y = 0$ (true border)	9.1 (2.1)	39.9 (13.0)
$y = -50$	-.9 (2.2)	4.4 (12.9)
$y = -75$	-1.4 (2.4)	2.3 (14.6)
$y = -100$	-3.3 (2.4)	-15.1 (14.8)

Table 6: Simulated-border placebo estimates.

6.6 Tables 7 and 8: Timing, robustness, and Fantus rankings

Read Table 7:

- The 1992 employment-share estimate is 6.6 in the baseline.
- Using manufacturing employment as a share of population, the discontinuity grows over time: 0.4 in 1947, 0.9 in 1954, 1.3 in 1963, 2.0 in 1987, and 2.5 in 1992.
- The small 1947 estimate suggests that the border gap was not already large at the start of the postwar period.
- Including western states produces a 1992 share estimate of 5.7, close to the baseline 6.6.

Read Table 8:

- The Fantus ranking orders states by a broad legislative business-climate index.
- Top-ranked states, such as Texas, Alabama, Virginia, the Dakotas, and the Carolinas, tend to have right-to-work laws.
- Low-ranked states, such as Massachusetts, California, and New York, do not.
- The high correlation between Fantus rankings and right-to-work status supports the idea that right-to-work is a proxy for a broader probusiness policy environment.

Economic message: The border gap develops over time and survives major sample checks. The Fantus ranking shows that the policy variation is broader than one labor law.

TABLE 7
ESTIMATES OF SHIFT PARAMETER θ FOR ALTERNATIVE
SPECIFICATIONS AND YEARS
(Excludes Coal Region)

	Estimate of θ	Observations
Share of 1992 employment (baseline case)	6.6 (1.6)	951
Share of population: 1992	2.5 (.6)	951
1987	2.0 (.5)	951
1982	1.8 (.6)	951
1972	1.2 (.6)	723
1963	1.3 (.5)	917
1954	.9 (.5)	901
1947	.4 (.4)	895
Manufacturing employ- ment growth: 1947–92 (baseline)	23.1 (9.2)	892
1963–92	13.9 (8.6)	915
1982–92	11.1 (6.0)	948
Includes western states: 1992 share	5.7 (1.3)	1,256
1947–92 growth	19.6 (9.2)	1,135

Table 7: Alternative years, outcomes, and
western-state robustness.

TABLE 8
1975 FANTUS LEGISLATIVE BUSINESS CLIMATE RANKINGS

State	1975 Fantus Ranking	Does State Have a Right-to-Work Law Now?
Texas	1	yes
Alabama	2	yes
Virginia	3	yes
South Dakota	4	yes
South Carolina	5	yes
North Carolina	6	yes
Florida	7	yes
Arkansas	8	yes
Indiana	9	no (had a law, repealed in 1965)
Utah	10	yes
North Dakota	11	yes
Mississippi	12	yes
Georgia	13	yes
Iowa	14	yes
Tennessee	15	yes
Arizona	16	yes
Nebraska	17	yes
Colorado	18	no
Missouri	19	no
Kansas	20	yes
Okahoma	21	no
Kentucky	22	no
New Mexico	23	no
Wyoming	24	yes
Idaho	25	yes (passed in 1985)
Louisiana	26	yes (passed in 1976)
Ohio	27	no
New Hampshire	28	no
West Virginia	29	no
Maine	30	no
Montana	31	no
Nevada	32	yes
Rhode Island	33	no
Wisconsin	34	no
Illinois	35	no
Maryland	36	no
New Jersey	37	no
Vermont	38	no
Washington	39	no
Oregon	40	no
Minnesota	41	no
Pennsylvania	42	no
Connecticut	43	no
Delaware	44	no
Michigan	45	no
Massachusetts	46	no
California	47	no
New York	48	no

SOURCE.—Weinstein and Firestone (1978).

Table 8: 1975 Fantus business climate rankings.

6.7 Tables 9 and 10: Alternative policy measure and large establishments

Read Table 9:

- The Fantus variable is strongly associated with manufacturing shares at right-to-work borders and at non-right-to-work borders.
- When all borders are used, the Fantus coefficient remains large and significant.
- When both Fantus and right-to-work status are included, the right-to-work coefficient shrinks substantially, while the Fantus coefficient remains important.

- This suggests that broader state business climate, not only right-to-work laws, is doing much of the explanatory work.

Read Table 10:

- Probusiness borders are associated with a higher share of employment in large establishments.
- For all industries, the right-to-work border effect is 6.64 percentage points when used alone.
- For manufacturing, the effect is much larger: 16.33 percentage points when right-to-work is used alone.
- For nonmanufacturing industries, the effect is small and insignificant.

Economic message: Broader state policy rankings predict manufacturing location even beyond right-to-work status. The large-establishment results fit the mechanism: probusiness policies appear especially relevant to large manufacturing plants.

TABLE 9
STATISTICAL MODEL FOR ALTERNATIVE POLICY VARIABLE

PARAMETER	JUST RIGHT-TO-WORK BORDERS (1)	JUST NON-RIGHT-TO-WORK BORDERS (2)	ALL BORDERS		
			(3)	(4)	(5)
A. County Manufacturing Shares					
ϕ (Fantus)	.38 (.08)	.32 (.09)	.35 (.06)	...	.28 (.07)
θ (right-to-work)	...	...	...	7.40 (1.47)	3.19 (1.85)
R^2	.516	.525	.546	.544	.547
Observations	923	1,837	2,760	2,760	2,760
B. County Growth Shares					
ϕ (Fantus)	2.28 (.53)	1.56 (.53)	1.96 (.37)	...	1.92 (.46)
θ (right-to-work)	...	...	...	31.03 (9.52)	1.46 (11.85)
R^2	.335	.362	.361	.356	.361
Observations	823	1,747	2,570	2,570	2,570

Table 9: Statistical model with the Fantus policy variable.

7 Short summary

- The paper asks whether probusiness state policies attract manufacturing.
- Right-to-work status is used as a simple proxy for a broader probusiness policy bundle.
- Simple state-level comparisons are confounded because right-to-work states differ geographically and historically from other states.
- Holmes instead studies state borders where right-to-work status changes.
- Counties are mapped by distance to the policy border and by position along the border.
- The main outcome is the 1992 manufacturing employment share; the second is 1947–92 manufacturing employment growth.
- Cross-tabulations show a large jump in manufacturing share and growth at the policy border.

TABLE 10
ESTIMATES OF STATISTICAL MODEL: SHARE OF EMPLOYMENT IN LARGE ESTABLISHMENTS

PARAMETER	ALL INDUSTRIES (N = 2,760)			MANUFACTURING (N = 2,728)			OTHER INDUSTRIES (N = 2,760)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ϕ (Fantus)	...	.24 (.07)	.12 (.09)	...	.60 (.12)	.33 (.15)	...	-.01 (.06)	.07 (.07)
θ (right-to-work)	6.64 (1.69)	...	4.76 (2.13)	16.33 (3.10)	...	11.33 (3.89)	1.55 (1.56)	...	2.65 (1.96)
R^2	.459	.458	.459	.434	.433	.435	.322	.322	.322

Table 10: Share of employment in large establishments.

- The main spatial regression estimates a 6.6-percentage-point jump in manufacturing share at the border.
- This is about one-third of the average manufacturing share.
- The growth model estimates a 23.1-point jump in postwar growth at the border.
- Simulated borders do not produce comparable jumps.
- The border gap is small in 1947 and grows over time.
- The results are robust to including western states.
- A broader Fantus business-climate ranking also predicts border discontinuities.
- Probusiness policies especially increase the employment share in large manufacturing establishments.
- The estimate is local and reflects a policy bundle; it is not a pure right-to-work law effect or a welfare estimate.

8 Conclusion

8.1 Core conclusion

Holmes provides strong evidence that state policy affects the location of manufacturing. Manufacturing activity rises discontinuously when one crosses from an antibusiness state into a probusiness state, and the estimated magnitude is economically large.

8.2 Economic intuition

Manufacturers care about policies that affect labor relations, taxes, regulation, and subsidies. When such policies differ sharply across neighboring states, firms near the border can choose the more favorable policy environment while changing little else about their location. This creates a local jump in manufacturing on the probusiness side.

8.3 Incremental contribution revisited

The paper's contribution is not simply to show that right-to-work states grew faster. That fact is already visible in aggregate data but is not causal. The contribution is the border design: by comparing nearby counties on opposite sides of the same state border, Holmes makes a much stronger case that policy differences matter.

8.4 Implications

The findings suggest that state-level economic policy can shape the geography of industry. They also imply that policies may redistribute manufacturing activity across states, especially near borders. The large-establishment evidence suggests that the policies are especially relevant for the kinds of plants most affected by unions, capital taxation, and regulation.

8.5 Limitations

The design is powerful but not perfect. State borders can coincide with natural or historical discontinuities. Right-to-work status is only a proxy for a larger policy bundle. The estimated effect is local to border areas and cannot be directly extrapolated to a statewide average. Finally, the paper cannot determine whether relocating manufacturing across state lines raises aggregate welfare.

8.6 Takeaway

The enduring takeaway is that state policies matter for manufacturing location, and that borders provide a credible way to see it. Holmes shows how geographic discontinuities can be used to separate policy effects from broad regional differences, while also reminding readers that the resulting estimates are local, policy-bundle effects rather than simple single-law causal effects.